

Day 9 — Design Your Own

Module: Magic Squares — Bhadragaṇita · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: create a personalised 4×4 or 5×5 magic square as visual art.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *kothumi vidhi* (IAST: kothumi vidhi; lit. “Nārāyaṇa’s algorithm for odd-order squares”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Personal magic-square design.** Each student designs ONE magic square. Options:
 - 3×3 with custom numbers (e.g. start from 5 instead of 1; magic constant changes accordingly).
 - 4×4 using Nārāyaṇa-style or Strachey method.
 - 5×5 using stair-step.
2. **Artistic treatment.** Calligraphy, colour-coding, geometric decoration. Inspired by mandala / yantra aesthetics or modernist grid art.
3. **Document the math.** Each student writes on the back: starting numbers, magic constant, method used, source reference. This is a “math behind the art” supplement.
4. **Source-anchoring.** Each student cites at least ONE source from this module (Nārāyaṇa, Khajuraho, Dürer, Plofker, etc.).
5. **Pair-critique.** Pairs swap; verify each other’s math correctness.

Activity (15 min)

A scaled version of *Class 3×3 Magic-Square Tournament* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Design Your Own.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).